# Assignment 3

Amani Hussien

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## 1 Part 1

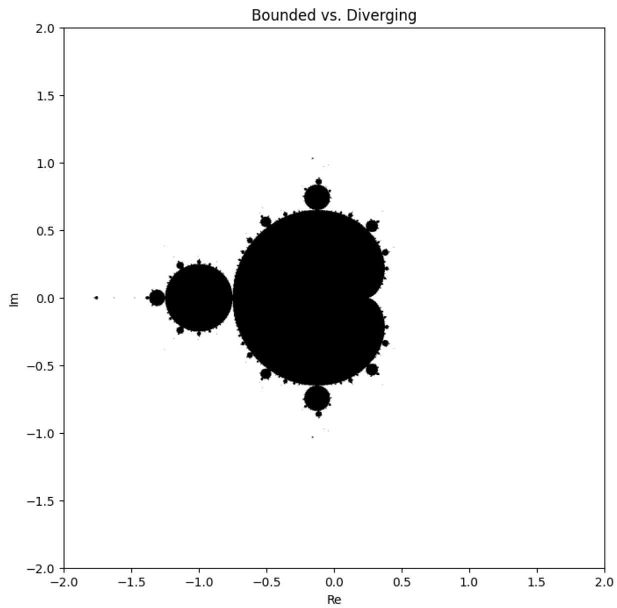
This program iterates through the recursive function

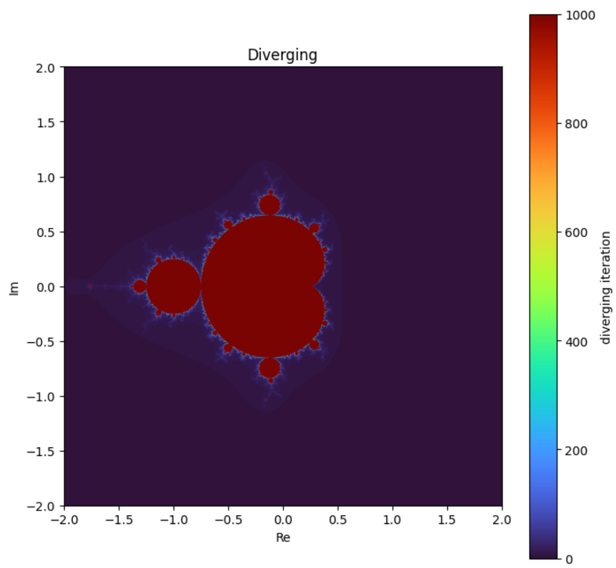
$$z_{n+1} = z_n^2 + c$$

over a grid of complex numbers $c = x + iy$ in where $-2 < x < 2$ and $-2 < y < 2$. The program tracks each point in the complex plane and determines whether the sequence is still bounded or has diverged. The following plots are, respectively:

1. Bounded vs Diverging plot: an image where points that are bounded are one color and points that have diverged are another

2. Divergence iteration plot: a color map showing the iteration number at which each point diverged




# 2 Part 2

This solves the Lorenz system of differential equations using SciPy's `solve_ivp` ODE solver. The Lorenz equations are given by

$$\frac{dx}{dt} = -\sigma(x - y)$$

$$\frac{dy}{dt} = rx - y - xz$$

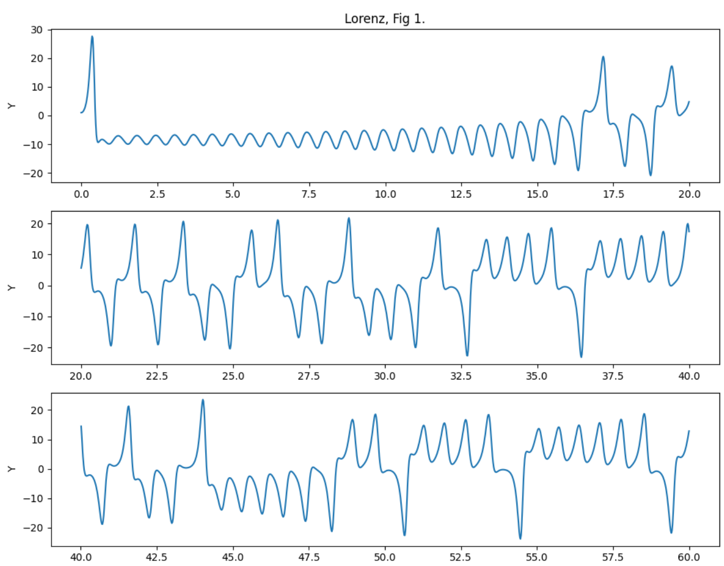
$$\frac{dz}{dt} = -bz + xy$$

where $\sigma$ is the Prandtl number, $r$ is the Rayleigh number, and $b$ is a dimensionless geometric parameter. The parameters used were

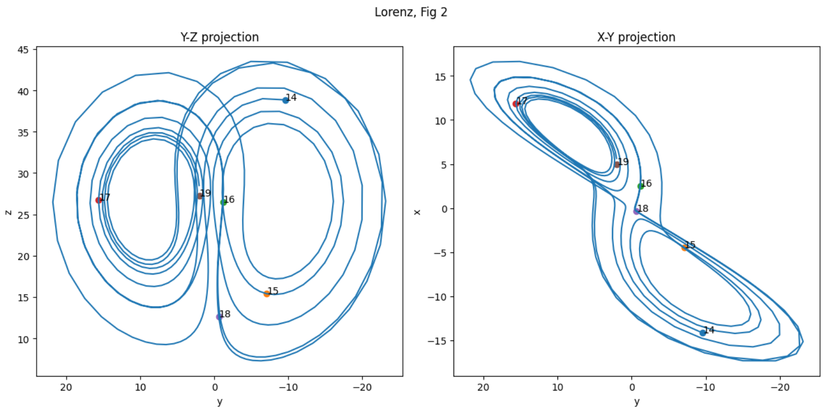
$$\sigma = 10, \qquad r = 28, \qquad b = \frac{8}{3}$$

with initial conditions $(x_0, y_0, z_0) = (0, 1, 0)$

The first figure shows the time-evolution of the y-coordinate over three consecutive time intervals. It is a reproduction of Fig 1. from Lorenz.




The second figure shows projections onto the $YZ$ and $XY$ planes. It is a reproduction of Fig 2. from Lorenz.




The last figure shows the system solved again with the initial condition differing by $10^{-8}$ in the $y$-component. The distance between these trajectories was then calculated as a function of time.